The Representer Theorem for Kernel Based Support Vector Machines

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Introduction

I will outline a proof of the representer theorem for reproducing kernel Hilbert spaces over $\mathbb{R}$, and then will use it in conjuction with the Support Vector Machine methods I have developed thus far to create a kernel based support vector machine.

Representer Theorem

Lemma 1: A Hilbert space $\mathcal{H}$ of functions from a set X to $\mathbb{R}$ (or $\mathbb{C}$) is a reproducing kernel Hilbert space if and only if $\exists K_x \in \mathcal{H}$ such that $\forall f \in \mathcal{H}$ $\langle f, K_x \rangle = f(x)$.

Proof. $\Longrightarrow$ By definition a Hilbert space H is a reproducing kernel Hilbert space if $\forall x \in X$ the linear functional $L_x : \mathcal{H} \to \mathbb{R}$ given by $L_x(f) = f(x)$ is continuous. Thus L_x is bounded on $\mathcal{H}$ as continuity implies boundedness, so we can apply Riesz's representation theorem to find a unique $g_x \in \mathcal{H}$ such that $L_x(f) = \langle f, g_x \rangle$. Setting $K_x = g_x$, it is clear that K_x is well defined as g_x is unique by Riesz's and $\langle f, K_x \rangle = \langle f, g_x \rangle = f(x)$. Thus, K_x is a reproducing kernel of $\mathcal{H}$.

$\Longleftarrow$ Suppose there exists a $K_x \in \mathcal{H}$ such that $\langle f, K_x \rangle = f(x)$. Then we can define a linear functional $L_x : X \to \mathbb{R}$ $L_x(f) = \langle f, K_x \rangle$. By Cauchy-Schwarz, $|\langle f, K_x \rangle| \leq ||f||_{\mathcal{H}}^2 ||K_x||_{\mathcal{H}}^2$. Thus, $L_x(f)$ is bounded. Now we must show that L_x is continuous. Given $\epsilon > 0$ pick $\delta = \frac{\epsilon}{||K_x||_{\mathcal{H}}^2}$. Then for $g \in \mathcal{H}$ such that $||f - g||_{\mathcal{H}}^2 < \delta$ we have:

$$
\begin{aligned}
||L_x(f) - L_x(g)|| &= |L_x(f - g)| \text{ as } L_x \text{ is a linear functional} \\
&= |\langle f - g, K_x \rangle| \\
&\text{By Cauchy Scharwz we have:} \\
&\leq ||f - g||_{\mathcal{H}}^2 ||K_x||_{\mathcal{H}}^2 \\
&< \frac{\epsilon}{||K_x||_{\mathcal{H}}^2} ||K_x||_{\mathcal{H}}^2 \\
&= \epsilon
\end{aligned}
$$

Thus, L_x is continuous and H is a reproducing kernel Hilbert space as required. □

Lemma 2: We can characterize K in the above lemma as a function $K : X \times X \to \mathbb{R}$ by $K(x, y) = \langle K_x, K_y \rangle$.

Proof. By Lemma 1, there exists $K_y \in \mathcal{H}$ such that $\forall f \in \mathcal{H}$ $\langle f, K_y \rangle = f(y)$. If we consider $K_x \in \mathcal{H}$, we can apply this characterization to it. We find that $K_x(y) = \langle K_x, K_y \rangle$. Thus we can define $K(x,y) = \langle K_x, K_y \rangle$ as a function from $X \times X$ to $\mathbb{R}$. In particular, this implies that K is symmetric. □

Theorem: Let $\mathcal{H}$ be a reproducing kernel Hilbert space of functions over $\mathbb{R}$ with domain V, a finite dimensional vector space over $\mathbb{R}$ and a set of inputs $X = \{x^{(1)}, \ldots, x^{(m)}\} \subset V$ and outputs $Y = \{y^{(1)}, \ldots, y^{(m)}\} \subset \mathbb{R}$. Now define $J : \mathcal{H} \to \mathbb{R}$ by $J(f) = \displaystyle\sum_{i=1}^{m} L(f(x^{(i)}), y^{(i)}) + \lambda ||f||_{\mathcal{H}}^2$ be a regularized risk function with respect to f. Then any function f^* that minimzes J has the form $f^* = \sum_{i=1}^{m} \alpha_i K(x, x^{(i)})$.

Proof. Given a reproducing kernel Hilbert space H with kernel K, it suffices to show that $\forall f \in \mathcal{H}$ we can find $f^* = \sum_{i=1}^{m} \alpha_i K(x, x^{(i)})$ such that $J(f) \geq J(f^*)$. Consider the subspace $W = \text{Span}\{K(\cdot, x^{(i)})\}_{i=1}^{m}$. W is finite dimensional by construction, so we may write $\mathcal{H} = W \oplus W^{\perp}$. Thus given $f \in \mathcal{H}$, we can find $f_1 \in W$ and $f_2 \in W^{\perp}$ such that $f = f_1 + f_2$. Applying the reproducing property of K to f_2 on $x^{(i)}$ we see that:

$$\begin{aligned}
f_2(x^{(i)}) &= \langle f_2, K_{x^{(i)}} \rangle \\
&\text{By Lemma 2:} \\
&= \langle f_2, K(x^{(i)}, \cdot) \rangle \\
&= \langle f_2, K(\cdot, x^{(i)}) \rangle \\
&\text{Since } f_2 \in W^{\perp}: \\
&= 0
\end{aligned}$$

Thus, $f(x_i) = f_1(x_i)$. Additionally by the pythagoras theorem: $||f||_{\mathcal{H}}^2 = ||f_1||_{\mathcal{H}}^2 + ||f_2||_{\mathcal{H}}^2 \geq ||f_1||_{\mathcal{H}}^2$ Now setting $f^* = f_1$. It is clear that:

$$\begin{aligned}
J(f) &= \sum_{i=1}^{m} L(f(x^{(i)}), y^{(i)}) + \lambda ||f||_{\mathcal{H}}^2 \\
&\geq \sum_{i=1}^{m} L(f^*(x^{(i)}), y^{(i)}) + \lambda ||f^*||_{\mathcal{H}}^2 \\
&= J(f^*)
\end{aligned}$$

□

Note: We assume L is convex in its first argument and lower semi-continuous to ensure it attains a minimizer in $\mathcal{H}$

The Kernel Trick

By the Moore-Aronszajn theorem, any symmetric and positive definite kernel K generates a unique reproducing kernel Hilbert space with K as its kernel. I want to revisit some of the computations in the HardMarginSVM.tex file. When we perform gradient descent we minimize a function D with respect to λ: $\displaystyle\sum_{i=1}^{n} \lambda_i - \frac{1}{2}||w||_2$, where w is also defined in terms of λ. Now, consider a feature mapping

ϕ which maps the inputs x_i to some different (possibly infinite dimensional) representation. We can replace all instances of x_i with $\phi(x_i)$ minimize in this new feature space. we previously had $w = \sum_{i=1}^{n} \lambda_i y_i x_i$ and so we now have $w = \sum_{i=1}^{n} \lambda_i y_i \phi(x_i)$. Furthermore, $||w||^2$ becomes:

$$\begin{aligned}
||w||^2 &= \langle w, w \rangle \\
&= \langle \sum_{i=1}^{n} \lambda_i y_i \phi(x_i), \sum_{i=1}^{n} \lambda_i y_i \phi(x_i) \rangle \\
&= \sum_{i=1}^{n} \sum_{j=1}^{n} \lambda_i \lambda_j y_i y_j \langle \phi(x_i), \phi(x_j) \rangle \\
&= \sum_{i=1}^{n} \sum_{j=1}^{n} \lambda_i \lambda_j y_i y_j K(x_i, x_j)
\end{aligned}$$

Note that K is the reproducing kernel induced by ϕ If we let $Y = yy^T$ where y is the vector formed of the labels, this is equivalent to:

$$||w||^2 = \lambda^T (Y \odot K) \lambda$$

Thus our problem becomes:

$$\min_{\lambda} (\lambda^T - \frac{1}{2} \lambda^T (Y \odot K) \lambda)$$

And we are guaranteed that the w we recover from the gradient identity because the representer theorem tells us that the optimal function that minimizes the loss is in the span of $\{K(\cdot, x^{(i)})\}_{i=1}^{m}$.